

A short parallel-query proof for fourfold forrelation

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Abstract

A single parallel quantum interrogation of four binary phase masks needs hard photon dose at least $(2/15)N^{1/8}$ to decide fourfold forrelation with error at most $1/3$, for powers of two $N \geq 2^{30}$. The proof has three parts: a matrix bound on Fourier coefficients, a correlated sign distribution, and concentration onto the exact promise. Arbitrary idlers, repeated modes, and coherence between photon-number sectors are allowed. The extension to outcome-dependent fresh batches is an open proof obligation; the earlier adaptive cut argument did not supply the required norm estimate.

1 Statement and proof outline

Let $N = 2^d$, with coordinates in $\mathbb{F}_2^d$, and set

$$H_{ij} = N^{-1/2}(-1)^{i \cdot j}, \quad u = N^{-1/2}(1, \dots, 1)^\top.$$

For four sign vectors $x^{(b)} \in \{\pm 1\}^N$ and $D_b = \text{diag}(x^{(b)})$, define

$$F(x) = \text{Forr}_4(x) = u^\top D_1 H D_2 H D_3 H D_4 u.$$

The task is to distinguish $F \geq 1/4$ from $F \leq -1/4$. A parallel probe is prepared before any sample output is available, passes through the direct-sum mask $D_1 \oplus \dots \oplus D_4$ once, and is measured arbitrarily. Hard dose D bounds the total number of signal photons on the entire support of the prepared state. References and idlers bypass the masks and are free.

Theorem 1. *For every power of two $N \geq 2^{30}$, a parallel probe solving the task with worst-case error at most $1/3$ has*

$$D \geq \frac{2}{15}N^{1/8}.$$

This includes mixtures of probes and predetermined collections of batches whose total hard dose is D : they can be prepared jointly and measured by a single final POVM.

We construct input laws μ_+, μ_- with mean forrelation approximately $\pm(2\rho/\pi)^3$, where $\rho = 200/201$. Their low-order sign correlations are small. Lemma 2 limits how strongly a parallel measurement can collect those correlations. Concentration then puts the laws on the required promises.

The proof below uses standard Gaussian Hermite expansion, the Gaussian arcsine identity, and the Efron–Stein and Gaussian Poincare inequalities. The arithmetic check `verify_constants.py` verifies the final constants, not these analytic steps. This package remains subject to independent review.

2 The parallel Fourier bound

Write $m = 4N$ for the total number of sign coordinates, and $\chi_S(x) = \prod_{a \in S} x_a$. A charge- D bosonic probe embeds in D labelled slots with alphabet $\{0, 1, \dots, m\}$, where 0 denotes vacuum. Symmetrizing a photon word with enough vacuum symbols gives an isometry from the truncated Fock space. The query is exactly $O_x^{\otimes D}$, with $O_x|0\rangle = |0\rangle$ and $O_x|a\rangle = x_a|a\rangle$. Enlarging to arbitrary slot states can only strengthen the competitor.

For a word $a = (a_1, \dots, a_D)$ let ∂a be the set of nonzero labels occurring oddly. Purify the input, and decompose it into orthogonal word components ψ_a . If $0 \leq E \leq I$ is the acceptance effect, then

$$f(x) = \sum_{a,b} \chi_{\partial a \triangle \partial b}(x) \langle \psi_a, E_{ab} \psi_b \rangle, \quad \sum_a \|\psi_a\|^2 = 1. \quad (1)$$

Thus $\hat{f}(S) = 0$ for $|S| > 2D$. Arbitrary idlers are included in the operator blocks E_{ab} . The Fourier coefficient is $\hat{f}(S) = 2^{-m} \sum_x f(x) \chi_S(x)$.

We use the elementary Schur factorization bound. If $K_{ab} = \langle p_a, q_b \rangle$ with $\|p_a\| \leq A$ and $\|q_b\| \leq B$, then

$$\|[K_{ab} E_{ab}]\| \leq AB \|E\|. \quad (2)$$

Map $(v_b)_b$ to $(q_b \otimes v_b)_b$, apply $I \otimes E$, and contract with the p_a to obtain (2). Let $\gamma_2(K)$ be the infimum of AB .

Lemma 2. *For arbitrary complex numbers c_S at level $1 \leq r \leq 2D$,*

$$\begin{aligned} \left| \sum_{|S|=r} c_S \hat{f}(S) \right| &\leq \max_{|S|=r} |c_S| \sum_{k=0}^r \binom{D}{k} \binom{D}{r-k} m^{\min(k, r-k)/2} \\ &\leq \binom{2D}{r} m^{r/4} \max_{|S|=r} |c_S|. \end{aligned} \quad (3)$$

Proof. In a pair of words (a, b) with $|\partial a \triangle \partial b| = r$, each differing label occurs oddly on exactly one side. Mark its last occurrence on that side. This gives a unique record of r marked positions among the D ket and D bra slots.

Fix a record with k ket marks. The marked labels are tuples $\alpha(a)$ of length k and $\beta(b)$ of length $r - k$. Let A, B be their underlying sets and put

$$S_a = \partial a \triangle A, \quad S_b = \partial b \triangle B.$$

A word is admissible on its own side if its marked labels are nonzero and distinct, each mark is the last occurrence of its label, and its residual parity is disjoint from its marked set. Write these two indicators as $\ell(a)$ and $r(b)$. Define

$$C_{\alpha\beta} = \begin{cases} c_{A \cup B}, & \text{all } r \text{ marked labels are distinct,} \\ 0, & \text{otherwise.} \end{cases}$$

The exact record contribution to (1) is the Schur symbol

$$K_{ab}^R = \ell(a) r(b) C_{\alpha(a), \beta(b)} \mathbf{1}_{\{S_a = S_b\}}. \quad (4)$$

Residual equality cancels every unmarked parity. The two disjointness conditions put each marked label on its specified odd side, and the last-occurrence conditions make the record unique. Repeated labels and vacuum slots are retained.

A factorization of C lifts to (4) by tensoring its row and column factors with $|S_a\rangle$ and $|S_b\rangle$, then multiplying by the indicators. Thus $\gamma_2(K^R) \leq \gamma_2(C)$. The matrix C has at most m^k rows and m^{r-k} columns. Standard basis vectors on its shorter side give

$$\gamma_2(C) \leq m^{\min(k, r-k)/2} \max |c_S|.$$

Apply (2) to (1), and sum over the $\binom{D}{k} \binom{D}{r-k}$ records. Vandermonde's identity finishes the proof. $\square$

3 Correlated signs with small moments

For $s = 1, 2, 3$, independently draw standard Gaussian vectors Z_s, W_s with cross-covariance $\mathbb{E}Z_s W_s^\top = \rho H$. This is a valid covariance because H is orthogonal and $\rho < 1$. Set $a_s = \text{sgn } Z_s$, $b_s = \text{sgn } W_s$ and form

$$y = (a_1, b_1 \odot a_2, b_2 \odot a_3, b_3).$$

Choose three independent uniform signs $\zeta_1, \zeta_2, \zeta_3$, put $\zeta_4 = \sigma \zeta_1 \zeta_2 \zeta_3$, and set $x^{(b)} = \zeta_b y^{(b)}$. This defines μ_σ .

For one link let $L(A, B) = \mathbb{E} \prod_{i \in A} a_i \prod_{j \in B} b_j$. If $a = |A|$, $b = |B|$ and $a + b \leq r \leq \sqrt{N}$, then

$$|L(A, B)| \leq \left(\frac{r}{2\sqrt{N}} \right)^{\max(a, b)}. \quad (5)$$

If one set is empty the moment is zero unless both are empty. Otherwise the cross-covariance submatrix has norm at most $\rho \sqrt{ab/N} \leq r/(2\sqrt{N})$. A product of a independent Gaussian signs has Hermite degree at least a . Conditional expectation on chaos degree p has norm at most $\|\rho H_{A,B}\|^p$. Pair equal degrees and apply Cauchy-Schwarz; both sign products have L^2 norm one. This proves (5).

For $S = (S_1, \dots, S_4)$ of total size r , gauge averaging kills the difference of its two moments unless every $q_b = |S_b|$ is odd. In that case

$$(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-})\chi_S = 2L(S_1, S_2)L(S_2, S_3)L(S_3, S_4).$$

Taking weights $(3/4, 1/4)$, $(1/2, 1/2)$ and $(1/4, 3/4)$ in the three maxima proves

$$\max(q_1, q_2) + \max(q_2, q_3) + \max(q_3, q_4) \geq 3r/4.$$

Consequently

$$|(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-})\chi_S| \leq 2^{1-3r/4} r^{3r/4} N^{-3r/8}. \quad (6)$$

No level below four contributes.

4 Concentration onto the promises

The arcsine identity gives $\mathbb{E}a_i b_j = \beta_N H_{ij}$, with

$$\beta_N = \frac{2\sqrt{N}}{\pi} \arcsin(\rho/\sqrt{N}).$$

The three links are independent. The defining $N^{-5/2}$ path-sum coefficient, three factors $N^{-1/2}$, and N^4 terms give

$$\mathbb{E}_{\mu_\sigma} F = \sigma \beta_N^3, \quad \beta_N^3 - \frac{1}{4} > \left(\frac{1400}{2211} \right)^3 - \frac{1}{4} > \frac{1}{260}. \quad (7)$$

Here $\arcsin z \geq z$ and $\pi < 22/7$ suffice.

Lemma 3. *For either sign law, $\text{Var } F < 1540/N$.*

Proof. Represent each link with independent standard Gaussians G, ξ, η :

$$a = \text{sgn}(G + \xi/\sqrt{200}), \quad b = \text{sgn}(HG + \eta/\sqrt{200}).$$

After standardization its correlation is ρH . Conditional on G_1, G_2, G_3 , all coordinates of the four ungauged blocks y are independent. Primitive conditional means are $\phi(G)$ and $\phi(HG)$, where

$$\phi(z) = \text{erf}(10z), \quad L = \|\phi'\|_\infty = 20/\sqrt{\pi}, \quad L^2 < 128.$$

The gauges multiply F by the deterministic sign σ .

For an independent biased sign Y , Efron–Stein bounds its variance contribution by $(1 - (\mathbb{E}Y)^2)(\partial_Y F)^2 \leq (\partial_Y F)^2$, where $\partial_Y F$ is the coefficient of Y . Each endpoint block has total squared derivative $1/N$. For block two the derivative is $\ell_j r_j$, with

$$\ell = HD_{a_1}u, \quad r = HD_{b_2 \odot a_3}HD_{b_3}u.$$

These vectors are independent, $\mathbb{E}\ell_j^2 = 1/N$, and $\|r\| = 1$. The contribution is $1/N$ after unconditional averaging. Block three is the same calculation with the ends reversed. Hence

$$\mathbb{E} \text{Var}(F \mid G_1, G_2, G_3) \leq 4/N. \quad (8)$$

Put $M(g) = D_{\phi(g)}HD_{\phi(Hg)}$, a contraction. The conditional mean is

$$\bar{F} = \sigma u^\top M(G_1)M(G_2)M(G_3)u.$$

At one link write this as $p^\top M(G)q$, up to σ . The partial-chain vectors p, q are independent of each other and G , with norms at most one and $\mathbb{E}|p_i|^2, \mathbb{E}|q_i|^2 \leq 1/N$. To justify the last statement, set $(P_s v)_i = v_{i+s}$ and $(D_s v)_i = (-1)^{s \cdot i} v_i$. The Sylvester identities and oddness of ϕ give

$$M(P_s g) = P_s M(g), \quad M(D_s g) = M(g)P_s.$$

Gaussian invariance makes the coordinate second moments of each nonempty left or right partial chain equal; the empty chain is u .

Differentiation, using $H = H^\top$, gives

$$\|\nabla_G \bar{F}\|^2 \leq 2L^2 (\|p \odot HD_{\phi(HG)}q\|^2 + \|q \odot HD_{\phi(G)}p\|^2).$$

Independence and the second-moment bounds make each expected squared norm at most $1/N$. Gaussian Poincare over the three links gives $\text{Var } \bar{F} \leq 12L^2/N$. With (8), $\text{Var } F \leq (4 + 12L^2)/N < 1540/N$. $\square$

5 Closing the bound

By (7) and Chebyshev,

$$\mu_\sigma\{\sigma F < 1/4\} \leq 104104000/N. \quad (9)$$

Conditioning on $\sigma F \geq 1/4$ changes each law in total variation by at most this probability. For $N \geq 2^{30}$ put $\varepsilon_0 = 1626625/16777216$. A correct acceptance probability $f \in [0, 1]$ must satisfy

$$\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f \geq \frac{1}{3} - 2\varepsilon_0 = \frac{3508733}{25165824}. \quad (10)$$

Suppose $D < (2/15)N^{1/8}$. Then $2D < \sqrt{N}$. Combine (3) and (6), recall $m = 4N$, and use $\binom{2D}{r} \leq (2eD/r)^r$:

$$\begin{aligned} |\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f| &\leq 2 \sum_{r=4}^{2D} \left(2^{3/4} e D r^{-1/4} N^{-1/8} \right)^r \\ &\leq \frac{2q^4}{1-q}, \quad q = 2eDN^{-1/8} 8^{-1/4} < \frac{11}{25}. \end{aligned}$$

The last inequality uses $e < 11/4$ and $8^{1/4} > 5/3$. The gap is less than $14641/109375$, whereas (10) is larger by

$$\frac{3508733}{25165824} - \frac{14641}{109375} = \frac{15314842691}{2752512000000} > 0.$$

This contradiction proves Theorem 1.

6 What remains for adaptive batches

For a fresh adaptive protocol, acceptance still has degree at most $2D$: multiply batch probabilities along each classical history. The Gaussian construction and concentration are unchanged. The additional requirement is precisely

$$\left| \sum_{|S|=r} c_S \hat{f}(S) \right| \leq \binom{2D}{r} (4N)^{r/4} \max_{|S|=r} |c_S| \quad (1 \leq r \leq 2D). \quad (11)$$

If (11) is proved for fresh adaptive protocols, the preceding proof establishes the original adaptive theorem with the same constants.

The parallel proof has a fixed input vector and a single effect $0 \leq E \leq I$ on its word space. An outcome-selected next probe does not supply that representation on aggregate words. The earlier fresh-cut proof asserted a common contraction after moving cuts through the adaptive network, but did not define its domains and maps sufficiently to establish that assertion. A product of contractions is contractive only with compatible domains, codomains, and tensor orientations; a general rearrangement of a tensor network is not such a product.

This is a gap in the supplied proof, not a counterexample to (11). The finite-size argument has a concrete norm-interface gap described in `../AUDIT.md`. Neither gap is resolved by checking constants. The present theorem concerns parallel interrogation; adaptive, mean-dose, and quantum-memory extensions are not claimed.